

Long division

This is a bit trickier. Make sure your brain is in gear before you tackle these problems!

Write $4081 \div 13$ like this:

$$13 \overline{)4081}$$

How many times does **13** go into **4**?
None, so move along one digit.

$$13 \overline{)4081} \quad \text{Work from left to right.}$$

How many times does **13** go into **40**? **3** times – so put a **3** above the **0**.

$$\begin{array}{r} 3 \\ 13 \overline{)4081} \\ \underline{-39} \\ 1 \end{array} \quad \begin{array}{l} 13 \text{ goes into } 40 \\ 3 \text{ times.} \\ 3 \times 13 = 39 \\ \text{Remainder} \end{array}$$

3 times **13** is only **39**. So we subtract **39** from **40** to get the remainder: **1**.

Now you need to deal with the **8**.
Move it down alongside the **1**, to make **18**.

$$\begin{array}{r} 31 \\ 13 \overline{)4081} \\ \underline{-39} \\ 18 \\ \underline{-13} \\ 5 \end{array} \quad \begin{array}{l} 13 \text{ goes into } 18 \\ \text{once.} \\ 1 \times 13 = 13 \\ \text{Remainder} \end{array}$$

How many times does **13** go into **18**? **1** time – so put a **1** above the **8**.

1 times **13** is **13**. Subtract this from **18** to find the remainder – **5**.

Move the **1** down.

How many times does **13** go into **51**?
3 times, with a remainder of **12**.

$$\begin{array}{r} 313 \\ 13 \overline{)4081} \\ \underline{-39} \\ 18 \\ \underline{-13} \\ 51 \\ \underline{-39} \\ 12 \end{array} \quad \begin{array}{l} 13 \text{ goes into } 51 \\ 3 \text{ times.} \\ 3 \times 13 = 39 \\ \text{Final remainder} \end{array}$$

There are no more digits to carry down, so we're finished. Phew.

The answer is **313**,
remainder **12**.



Now have
a go!

$$36 \overline{)845}$$

$$24 \overline{)5361}$$

$$13 \overline{)823}$$

$$23 \overline{)4810}$$

$$18 \overline{)417}$$

I think I need a
break now!

